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Online Gaming Network Communication Dynamics, Depressive Symptoms, and Social Support: A Longitudinal Network Analysis

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ABSTRACT

Video games are part of everyday life for many Americans despite concerns for social isolation and depressive symptoms. Preliminary studies show gamers may compensate for lack of in-real-life (IRL) support with online connections. This longitudinal social network study investigated the social structure of an online gaming site and how social support, sense of community, and depressive symptoms relate to communication. Members (N = 40) of an online gaming site reported online and IRL support, sense of community, depressive symptoms, and usernames of other members whom they spoke to about important life matters. IRL and online social support, sense of community, and depressive symptoms significantly influenced changes in online gaming network structure over time. These results are timely given social isolation and mental health impacts related to the COVID-19 pandemic. Exploring how to healthfully build online connections through gaming may be an avenue for greater social support when IRL social support is lacking.

Video games are a part of everyday life for many Americans. It is estimated that over 65% of Americans play some form of video game daily, while the video game industry accounted for over \$43.4 billion in revenue in 2018 (ESA 2019). This increase in use and popularity has been met with many concerns over its potential negative health effects, including risk for isolation, addiction, increased aggressive behavior, and increased risk for depressive symptoms (Kowert and Quandt 2015; Wei et al. 2012). Excessive internet use is also associated with increased risk for depression and anxiety (Christakis et al. 2011; Kowert and Quandt 2015). Furthermore, “Internet Gaming Disorder” was included in the fifth edition of the *Diagnostic and Statistical Manual of Mental Disorders (DSM-5)* in 2013 as one of the “Conditions for Further Study” (APA 2013). This possible disorder includes loss of interest in activities and use of games to “escape negative mood” when the gaming is considered to be at a “problematic level” (i.e., disrupts normal life functioning or becomes compulsive) (APA 2013). However, it should be noted that these criteria have been criticized for possibly overpathologizing normal game behavior (King et al. 2020).

The World Health Organization (WHO) identifies depression as a leading cause of disability worldwide, with over 300 million people suffering from depression globally (Murray et al. 2013; WHO 2021). One in ten American adults currently suffers from a form of depression, and one in five will experience it at least once in their lifetime (Brody, Pratt, and Hughes 2018). Most notably, this figure has significantly increased over the past decade for individuals between the ages of 17 and 25 (Weinberger et al. 2018). Symptoms of depression (which can vary in severity and length) include, but are not limited to, depressed mood, diminished interest in activities that once brought pleasure, significant changes in weight or appetite,

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issues with sleep, slow or agitated motor functioning, fatigue, feelings of worthlessness or guilt, issues with concentration or focus, and thoughts of death or dying (APA 2013). Despite the prevalence and treatability of depressive symptoms, many do struggle to seek professional help (Schnyder et al. 2017). Moreover, when individuals encounter difficult situations, they tend to turn to their social support networks for guidance and coping (Small 2017). Hence, investigating these networks is pertinent.

The network-episode model (NEM) is a framework used to understand the attitudes and actions of a person considering seeking help for mental health issues through both formal and informal social networks (Pescosolido, Gardner, and Lubell 1998). This model addresses the continual thoughts and experiences involved with the decision-making process and how it is constantly influenced by social interactions and environmental context (Pescosolido, Gardner, and Lubell 1998). The NEM asserts that health problems, specifically severe problems that exceed the individual's capacity to cope, force people to activate both formal and informal social networks as a means to manage their problems (Perry and Pescosolido 2015). Due to negative stigma associated with mental health services (Clement et al. 2015; Vogel et al. 2014), research suggests that informal sources of support, specifically through online communities, may serve an important role in help-seeking and coping behaviors (Heerde and Hemphill 2018). Many online games provide a chat function or forums for players to extend real-life relationships and make new online friendships, which may facilitate this informal help-seeking behavior (Trepte, Reinecke, and Juechems 2012). Furthermore, a longitudinal study concluded that young adults may use online games to compensate for preexisting in-person social difficulties (Kowert et al. 2015). In addition, coping through gaming is suggested to be important for those who report less in-real-life (IRL) social support (Reinecke 2009). However, gaming can be a source of joy, coping, and support for all regardless of feelings of IRL social support (Kowert and Quandt 2015; Raith et al. 2021; Trepte, Reinecke, and Juechems 2012).

These social connections present online can be examined using social network analysis (SNA). SNA is a set of theories and methods that allow researchers to investigate connections between individuals and analyze the social structure of groups (Borgatti, Everett, and Johnson 2018). In this manner, researchers can examine the underlying social structure, influence of social connections throughout a group of individuals, and the impacts on their health (Valente 2010). Specifically, whole network (or sociocentric) network analysis aims to understand the social connections within a bounded group of individuals (e.g., members of an online gaming community) and how these connections impact outcomes individually and collectively. In cross-sectional pilot studies using SNA to investigate online gaming communities, members were similar in depressive symptoms scores to those they spoke with online, and reduced IRL support was associated with more online connections (Prochnow et al. 2020, 2020).

Study Purpose

While existing literature suggests the importance of network connectivity and support for online gamers, these effects have only been observed cross-sectionally. Due to evidence that online relationship building within gaming environments is linked to depression, particularly for those who experience greater social isolation in real life, it is important to understand how the association between network position, social support, and depressive symptoms changes over time. Thus, this study aims to use SNA to investigate the social structure of an online gaming site across two time points to examine how factors such as social support, sense of community, and depressive symptoms impact social connections over time. Subsequently, this pilot study aims to use longitudinal network modeling procedures, which (to the authors' knowledge) have not yet been used to understand online gaming network communication. This serves as a vital first step toward the utilization of these techniques in larger gaming communities.

Methods

This study examined a specific network of online gamers (i.e., members of the same gaming site) over a 10-month time frame. This gaming site was chosen because of the size ($N = 101$) and the posting of a clear member list (i.e., a list of all the people involved in the gaming site), which allowed the ability to perform sociocentric network analysis. Members were classified as “active” if they had logged in to the gaming site within the last two weeks. This site is a football simulation site where members performed the role of head coach and competed against other members weekly over a 10-month period. Site members can communicate with other members directly on the site through forums, chat functions, and direct messaging. By posting on the site’s main forum, all active members over the age of 18 were invited to participate in an online survey. Members were surveyed in May of 2019 and again in March of 2020, at the start and finish, respectively, of the online game’s season. All study procedures were approved by the Institutional Review Board (IRB) of the referent university prior to the start of the study.

Measures

Online gaming site member characteristics – including age, sex, race, ethnicity, educational attainment, and marital status – were collected. Members were also asked to report the number of hours per week they spent on the gaming site and the number of hours they spent playing other video games.

Depressive Symptoms

The 9-item Patient Health Questionnaire (PHQ-9) was used to measure depressive symptoms (Kurt, Spitzer, and Williams 2001). With this scale, members were asked the frequency with which they had been bothered with certain problems over the last two weeks – ranging from “not at all,” “several days,” “more than half the days,” to “nearly every day.” Responses were then scored from zero to three, respectively, and subsequently summed to create a total scale score. Example problem items include “Little interest or pleasure in doing things,” “Feeling down, depressed, or hopeless,” and “Feeling bad about yourself – or that you are a failure or have let yourself or family down.” Recent literature reviews suggest that a score of 14 or greater demonstrates the possibility of severe clinical depression (Kurt, Spitzer, and Williams 2001). While results from the questionnaire should be verified by a clinician, two meta-analyses concluded this cutoff to have acceptable diagnostic properties for detecting depressive episodes (Manea, Gilbody, and McMillan 2012, 2015). This scale has been previously validated in a sample of primary care patients and has good internal ($\alpha = 0.89$) and test-retest (Pearson’s $r = 0.84$) reliability, as well as diagnostic validity (sensitivity: 88%, specificity: 88%) (Kurt, Spitzer, and Williams 2001). There was good internal reliability within this study’s sample ($\alpha = 0.87$) (Cronbach 1951).

Appraisal Social Support

This study used a form of social support defined as “appraisal” social support, or a perceived availability to talk to someone about one’s problems, which was measured by an adapted version of the Interpersonal Support Evaluation List (ISEL-12) (Sheldon et al. 1985). The original scale included three subscales (i.e., belonging, tangible, and appraisal); additional items from the belonging and tangible subscales were not included in the present study. Sample items are “I feel that there is someone I can share my most private worries and fears with” and “There is someone I can turn to for advice about handling problems with my family.” Participants were asked if they strongly disagree, disagree, agree, or strongly agree with each statement, which were scored one to four. Scores were then averaged for an overall mean scale score. Participants were first asked to think about IRL people and answer each item. Then, participants were asked to consider only their friends on the gaming site and answer each item. Scores for the IRL and online gaming site items were averaged separately to generate scale scores for each setting. In this sample, both online gaming and IRL scales displayed fair to good internal reliability ($\alpha = 0.85, 0.74$, respectively) (Cronbach 1951).

Sense of Community

Sense of community was assessed in this study using an adapted scale first published by Conn (2017). Sense of community is important to understand within online gaming sites since previous research has examined how online communities build a vital “third place” (i.e., a place outside work and home where an individual feels comfortable) for individuals to connect and share with others (Steinkuehler and Williams 2006). Sense of community was measured using a four-item scale with response options of strongly disagree, disagree, agree, and strongly agree. Items include “I feel like I belong here,” “Being a member here fills an important need in my life,” “I feel proud to be a part of this community,” and “There is a strong sense of community on this site.” Responses were coded one to four and then averaged to obtain a composite score. The scale displayed fair internal reliability in this sample ($\alpha = 0.75$) (Cronbach 1951).

Social network data.

To generate a network for analyses, members of the online gaming site were given a list of all active members on the site (i.e., a member roster) and asked to select any and all members whom they speak to about important life matters. This name generator is frequently used to obtain a list of social connections (Adams 2019). Participants could select as many other members outside of themselves (i.e., alters) as they would like by checking the box for each person within the survey. The list of names included in the survey was updated for the second time point to include any changes to the network since the first round of data collection.

Data Analysis

Summary statistics – including means, standard deviations, and frequencies – were calculated for member demographics. T-tests were calculated for support and depressive symptoms scores between time points. Means, standard deviations, frequencies and t-tests were all conducted using SPSS version 25 (IBM 2018). Additionally, a Jaccard index was computed on the network data to determine whether connections were maintained, created, lost, or remained absent from 2019 to 2020. Participants who responded to both surveys were then included in longitudinal network modeling.

Separable temporal exponential random graph models (STERGMs) were used to model changes in the network over time (Krivitsky and Handcock 2014). STERGMs are an extension of exponential random graph modeling, which is used to determine significant factors associated with the presence of a connection in cross-sectional networks (Lusher, Koskinen, and Robins 2013). Separable in this instance means that the formation and maintenance of connections are modeled individually in two separate models (Krivitsky and Handcock 2014). STERGMs use iterative Markov chain Monte Carlo (MCMC) algorithms to approximate the maximum likelihood estimates for the log-odds of associations between given parameters (e.g., factors related to network structure or characteristics of the individuals in the network) and tie formation or tie maintenance (Lusher, Koskinen, and Robins 2013). This modeling procedure simulates networks based on empirical data and parameters set by the researcher. STERGMs determined the probability of certain network properties occurring at random within each network (Lusher, Koskinen, and Robins 2013). For example, the researcher could choose to test whether a gamer who reported higher IRL social support scores was more or less likely to send or receive a tie in their network over time. Significant parameter estimates indicate that the specific network configuration is observed in the empirical network more than would be expected by chance, given the other effects modeled in the network. Parameter estimates (PEs) and standard errors (SEs) were provided for each factor entered in the model. A positive significant parameter in the formation model would suggest that the given parameter configuration was more likely to form from time one to time two. Conversely, a negative parameter would suggest less likelihood of the configuration forming over time. A positive significant parameter in the dissolution model would suggest that the given parameter configuration was more likely to be *maintained* between time points. On the contrary, a negative significant parameter in the dissolution model would suggest that the given parameter configuration was less likely to be maintained between time points. Model parameters were entered

into the model and then checked for degeneracy or aspects that might cause irregular network data. Models showed adequate fit using MCMC diagnostics. Cleaning and management of network data, as well as STERGM analyses, were done using statnet package in R Studio (Hunter et al. 2008).

Model Specification

Network structure parameters were added to model density (i.e., edges or connections in the network), reciprocated connections (i.e., connections that were mutually shared between two individuals), and transitive triads (i.e., three individuals who were connected to each other). A parameter related to homophily based on depressive symptom score was added using an absolute difference parameter to understand if members tended to connect with others who were similar to them in depressive symptoms score. Sender and receiver covariates were added to determine if time spent on the site, depressive symptoms, online social support, IRL social support, sense of community, relationship status, and educational attainment were related to someone sending or receiving communication ties in the network.

Results

Sample Characteristics

In total, 37 members (37% of total active membership) responded at time one, and 40 members (40% of active membership) responded at time two—30 of which were represented in both timepoints. Respondents were, on average, 24.76 years old ($SD = 6.55$) at time one and 25.25 years old ($SD = 6.88$) at time two. All respondents were male at both time points, and 89.2% and 80% indicated that they were white at each respective time point. Most of the respondents indicated that they were employed (64.9 percent, 70 percent), while roughly a third (35.1%, 30%) identified as students. Most of the sample had a bachelor's degree or beyond. See Table 1 for the complete sample characteristics at both timepoints.

There was no significant difference between the mean PHQ-9 scores of respondents at time one ($M = 6.92$, $SD = 5.86$) and time two ($M = 5.85$, $SD = 4.68$): $t(75) = 1.06$, $p = .29$. Five members (13.5%) reported a score greater than 14 at time one, which is the cutoff point for identifying the possibility of severe clinical depression (Manea, Gilbody, and McMillan 2012, 2015). Subsequently, only three members (7.5%) met this threshold at time two. Similarly, there were no differences between time one and time two for hours spent on the site ($t(75) = 0.62$, $p = .54$), IRL support ($t(75) = 0.04$, $p = .97$), online support ($t(75) = 1.01$, $p = .32$), and sense of community ($t(75) = 0.95$, $p = .35$). Means and standard deviations for all measures can be found in Table 1.

Stergm

In total, 101 connections were reported at time point one. Over time, 66 of those connections were maintained, 45 were dissolved, and 128 new connections were formed – resulting in 194 at time point two. This change resulted in a Jaccard Index of 0.28.

Significant factors for formation and maintenance of communication ties were modeled. Connections were significantly more likely to form if the connection was part of a reciprocal ($PE = 0.87$, $SE = 0.30$) or transitive relationship ($PE = 0.71$, $SE = 0.12$). Furthermore, communication ties were significantly more likely to be maintained if they were transitive ($PE = 0.56$, $SE = 0.27$). Members were significantly more likely to form ($PE = 0.07$, $SE = 0.02$) and maintain ($PE = 0.17$, $SE = 0.08$) communication ties with members who were dissimilar to them in age. Older members were significantly more likely to receive ($PE = 0.03$, $SE = 0.01$) and send ($PE = 0.04$, $SE = 0.02$) more communication ties over time as compared to younger members. Meanwhile, members were significantly more likely to send more communication ties over time if they reported lower IRL support ($PE = -0.52$, $SE = 0.18$), more online support ($PE = 0.44$, $SE = 0.16$), greater sense of community ($PE = 0.38$,

Table 1. Sample characteristics.

	Time 1 (N = 37)		Time 2 (N = 40)	
	n (%)	Mean (SD)	n (%)	Mean (SD)
Race				
Asian	2 (5.4%)		2 (5.0%)	
Black	2 (5.4%)		6 (15.0%)	
White	33 (89.2%)		32 (80.0%)	
Ethnicity				
Hispanic	3 (8.1%)		1 (2.5%)	
Non-Hispanic	34 (91.9%)		39 (97.5%)	
Education				
High School/GED	13 (35.1%)		14 (35.0%)	
Some College	3 (8.1%)		1 (2.5%)	
Bachelor's Degree	17 (45.9%)		19 (47.5%)	
Graduate Degree	4 (10.8%)		6 (15.0%)	
Employment				
Employed	24 (64.9%)		28 (70.0%)	
Student	13 (35.1%)		12 (30.0%)	
Marital Status				
Single (Not married)	30 (81.1%)		34 (85.0%)	
Married	7 (18.9%)		6 (15.0%)	
Age		24.76 (6.55)		25.25 (6.88)
Hours on Gaming Site		12.57 (8.60)		14.75 (12.38)
Other Gaming Hours		12.05 (9.55)		13.28 (12.68)
IRL Support		3.49 (0.57)		3.45 (0.90)
Online Support		2.81 (0.84)		2.98 (0.68)
Sense of Community		3.21 (0.58)		3.33 (0.45)
Depressive Symptoms		6.92 (5.86)		5.85 (4.68)

SE = 0.19), and fewer depressive symptoms (PE = -0.06, SE = 0.02). Additionally, members who reported more IRL support were significantly less likely to maintain outgoing communication ties over time (PE = -1.19, SE = 0.58). Members who spent more hours on the site were significantly less likely to maintain incoming communication ties over time (PE = -0.06, SE = 0.02). [Table 2](#) presents the complete STERGM modeling parameters and results.

Discussion

This study explored factors related to communication ties within an online gaming network over time. Until this point, there has been a dearth of literature investigating online gaming sites and their implications for depressive symptoms using longitudinal network modeling. Results suggest social support, sense of community, and depressive symptoms significantly influenced changes in the online gaming network structure overtime.

Age was a significant factor in the formation of communication ties in this network. Specifically, communication ties were significantly more likely to form between members of different ages. This may suggest younger members are seeking out older members for advice or guidance. On the contrary, older members may be connecting with younger members to bridge technology gaps (Venter 2017) or to feel more like a part of the gaming community, which typically trends younger (Rudolf et al. 2020). Furthermore, older members were also significantly more likely to form communication ties over time (both sending and receiving) compared to younger members. This could be a reflection of these members filling a social connection gap after collegiate years or as members age. This result also suggests that these members may be the driving force behind the communication that occurs in this network.

Structural factors such as transitivity and reciprocity were significant in the formation of ties over time. In other words, communication ties within this network were significantly more likely to form if

they were reciprocated or if they were a part of a transitive relationship (i.e., a common third member whom both members were connected to). In-person communication and network structures tend to follow this pattern of strong ties and triadic closure (Kammrath et al. 2020). Similarly, communication ties were significantly more likely to be maintained over time between members who reported more hours spent on the site each week. Just like IRL relationships, online relationships require time to maintain (Rivera, Soderstrom, and Uzzi 2010). More hours spent on the site would mean more opportunities to chat, connect, and strengthen relationships with other members (Trepte, Reinecke, and Juechems 2012). Naturally, this extended time and these stronger relationships can lead to greater feelings of social support as well (Trepte, Reinecke, and Juechems 2012).

Members who reported feeling a greater sense of community were also more likely to form ties over time. Members who feel a sense of pride in the site and value being a part of it would potentially be more likely to make more connections and feel more comfortable conversing with other members about important life matters. This type of comfort through sense of community is noted within college students, as students who reported more sense of community tended to make more friends and had a greater chance of remaining at their institution (Conn 2017). Cultivating a sense of community within online gaming sites can be a vital way for developers to engage and connect gamers (Arbeit et al. 2016; Trepte, Reinecke, and Juechems 2012). Future studies should investigate the elements within a game or interface that might foster these feelings of community. However, it should also be noted that connecting with likeminded others and building a strong sense of community in this way may also come with negative unintended (or intended) consequences – such as toxic masculinity, political or ideological echo chambers, and others (Fabian et al. 2020; Haase 2021).

Furthermore, members were significantly more likely to form outgoing ties over time if they reported more online support. In other words, participants who reported receiving more social support online were more likely to form outgoing ties within their network overtime. This result supports the NEM framework, suggesting people find informal support channels through various social networks, including online communities (Perry and Pescosolido 2015). Research also suggests that the more supported a person feels, the more likely and comfortable they are expanding their network and engaging outside of their comfort zone (Jungwon et al. 2017; Perry and Pescosolido 2015). In this case, that could mean initiating new social connections across the gaming platform. Developing contact points for these members to communicate about the game, as well as other important life matters, may help provide greater feelings of support for these individuals and could buffer people from experiencing depression and other mental health issues (Arbeit et al. 2016; Trepte, Reinecke, and Juechems 2012; Wang et al. 2018).

Not only were those that reported more online support likely to create ties over time, but it was also those who reported less IRL support who created more ties, and dissolved less ties, over time. This finding is congruent with previous research suggesting that people with less developed support systems in real life tend to find more meaningful social connections online (Kowert et al. 2015). Past literature reported that young adults may use online means as a way to adjust for preexisting in-person social difficulties (Kowert et al. 2015). This finding also supports previous pilot studies hypothesizing that members who felt less IRL social support may be compensating for this deficit by reaching out within their online gaming community (Prochnow et al. 2020; Tyler, Patterson, and Hartnell 2020). Thus, online gaming networks may be an important opportunity to grow social networks for people who have less opportunity to do so in real life.

Despite the above noted effects of social support and sense of community, members were significantly less likely to send communication ties over time if they reported more depressive symptoms. This aligns with studies associating depression with reduced social connection and social isolation (Wang et al. 2018). However, this could mean that those with higher depression scores need more formal or IRL support systems as compared to those with lower scores (Gough 2016), since new ties created online may not provide the support needed. That said, while the gaming site may be a place to bolster a lack of IRL social support using online connections for those with lower depressive scores, it may not be ideal for those with higher

Table 2. Final model parameter estimates and standard errors for communication tie formation and dissolution among gamers.

Parameters	Formation		Maintenance	
	PE (SE)	<i>p</i> -value	PE (SE)	<i>p</i> -value
Edges	−5.72 (1.26)	<0.01*	6.61 (5.31)	0.21
Reciprocity	0.87 (0.30)	<0.01*	−0.17 (0.74)	0.80
Transitivity ^x	0.71 (0.12)	<0.01*	0.56 (0.27)	0.04*
Homophily ^y				
Age	0.07 (0.02)	<0.01*	0.17 (0.08)	0.03*
Depressive Symptoms	0.04 (0.03)	0.15	−0.10 (0.06)	0.14
Receiver Covariates				
Age	0.03 (0.01)	0.02*	−0.10 (0.07)	0.13
Sense of Community	−0.12 (0.19)	0.53	0.60 (0.59)	0.30
Site Hours	−0.02 (0.01)	0.06	0.06 (0.02)	0.02*
IRL Support	−0.01 (0.18)	0.94	−0.32 (0.50)	0.52
Online Support	0.05 (0.15)	0.72	0.49 (0.51)	0.33
Depressive Symptoms	0.01 (0.02)	0.88	0.06 (0.07)	0.38
Sender Covariates				
Age	0.04 (0.02)	<0.01*	0.02 (0.06)	0.74
Sense of Community	0.38 (0.19)	0.04*	0.16 (0.43)	0.71
Site Hours	0.01 (0.01)	0.18	0.01 (0.02)	0.42
IRL Support	−0.52 (0.18)	<0.01*	−1.19 (0.58)	0.04*
Online Support	0.44 (0.16)	<0.01*	−0.62 (0.37)	0.09
Depressive Symptoms	−0.06 (0.02)	0.02*	−0.02 (0.05)	0.63

Note: PE – parameter estimate, SE – standard error.

^xTransitivity was modeled using geometrically weighted edgewise shared partner distribution with a decay of 0.10.

*Indicates a significant effect at $p < .05$.

^yHomophily is measured using the absolute difference parameter. This means positive PEs would represent a predisposition to connect with others who differ more on a given characteristic (e.g., age).

depression scores. Furthermore, based on evidence that depression can lead to social isolation, the online gaming community may not be resistant to the negative isolating implications of depressive symptoms (Wang et al. 2018). This is especially concerning given social isolation and reduced social networks can exacerbate depressive symptoms, creating a harmful cycle of depressive symptoms and loneliness (Wang et al. 2018).

Limitations

The main limitations to this study are the sample size and specificity to the specific gaming site. While the size of the network is a limitation, the analyses here use simulations to provide adequate power. The results here, while descriptive of a small online gaming community, are not robust enough to be generalized across other sites and other gaming genres. While there are many sports simulation games similar to the one seen here, the generalizability to other games should be cautioned against. This gaming site was chosen specifically to be a pilot, as it provided a convenient membership list and provided opportunities for the members to communicate regularly. The size also allowed all members the opportunity to interact with all other members, which is not the case in all games yet is a stipulation of the statistical analysis conducted. The investigation of online gaming communication and impacts on depressive symptoms and social support will need to be replicated in other games, genres, and modalities of gaming. Moreover, the order of questions was not randomized, which could bias the measurement of IRL and online social support. Also, the lack of a specific timeframe for connection within the network-generating question could introduce bias in the addition or dissolution of contacts. Additionally, the sample was all male and largely white, non-Hispanic males, which limits generalizability.

Implications

This study supports the NEM framework and adds to a body of literature suggesting that online social connections can yield support. Furthermore, it provides evidence that these effects are most salient for people who are less supported in real life. Findings suggest potential in online intervention strategies to increase support for populations who are traditionally more socially isolated (e.g., rural populations, older adults, and deployed soldiers) and could benefit from supportive connections created through digital platforms (de Koning, Richards, and Stathi 2020; Gemma, Hill, and Kiernan 2018). Additionally, the use of STERGMs in this study provides a vital first step into using longitudinal network modeling in the investigation of these support structures.

These results are especially timely and important given the increases in social isolation and mental health impacts related to the COVID-19 pandemic. Given many people experienced reduced levels of IRL support due to the pandemic (Saltzman, Hansel, and Bordnick 2020), this study suggests that online connectivity could be a remedy during more socially isolating times. However, because these results suggest that these effects do not seem to hold for those most vulnerable to depressive symptoms, improving mental healthcare access should be a priority, during and following COVID-19, especially due to increases in depression as a result of the pandemic (Saltzman, Hansel, and Bordnick 2020). Based on results from this study, combined with trends related to COVID-19, there is greater rationale for promoting alternative solutions to in-person therapy such as telehealth strategies to reduce mental-health-related burden, especially for medically underserved communities and potentially for online gamers who experience depression (Hilty et al. 2013; Tarlow et al. 2014).

Ultimately, this study reaffirmed that the relationships between mental health, social support, and online gaming are complex. More research is needed to determine the relationship between gaming and depressive symptoms, with the potential for social networks to mediate the relationship between the two. Future experimental studies could investigate if fostering relationships within online gaming communities, and therefore increasing networks and support systems for those who register higher depressive scores, reduces depressive symptoms and potentially serves as a buffer for mental health promotion. Egocentric network studies could also determine if a combination of formal and informal ties for those with depressive symptoms, as indicated by the NEM, might improve the social experience of online gamers over time. Perhaps if they received the IRL support they need via formal healthcare, they could experience a greater sense of inclusion among their gaming networks.

Conclusions

Online gaming provides an opportunity for individuals to connect and communicate. Examining the impact of this communication on feelings of support and depressive symptoms is increasingly important as the popularity of gaming increases. This study provides preliminary evidence of these implications using longitudinal network modeling (i.e., STERGM). More studies should employ longitudinal modeling to evaluate network dynamics. Furthermore, social support, sense of community, and depressive symptoms significantly influenced dynamic changes in the online gaming network communication structure. Future studies may wish to employ mixed methods or qualitative approaches to further investigate these associations.

Disclosure Statement

No potential conflict of interest was reported by the authors.

Notes on contributors

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